

WASP-39b

R-1640

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_160316 · created 2026-08-02T06:47:02+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457463.8795 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.01
Transit depth (Rp/Rs)^2	2.86 +/- 0.35 [%]
Orbital Inclination (inc)	86.6 +/- 0.4
Airmass coefficient 1 (a1)	1.21 +/- 0.012
Airmass coefficient 2 (a2)	-0.1394 +/- 0.0038
Scatter in the residuals of the lightcurve fit is	0.96 %
Best Comparison Star	None
Optimal Aperture	3.85
Optimal Annulus	7.59
Transit Duration (day)	0.1079 +/- 0.0063

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.169 $\pm$ 0.01 vs 0.1457 (deviation 1.76 σ)
Duration fit vs published	0.1079 d vs 0.12 d (deviation 1.45 σ)
Mid-time O-C	-2.4 min
Depth SNR	12.8
Residual scatter	0.96 %

Light curve

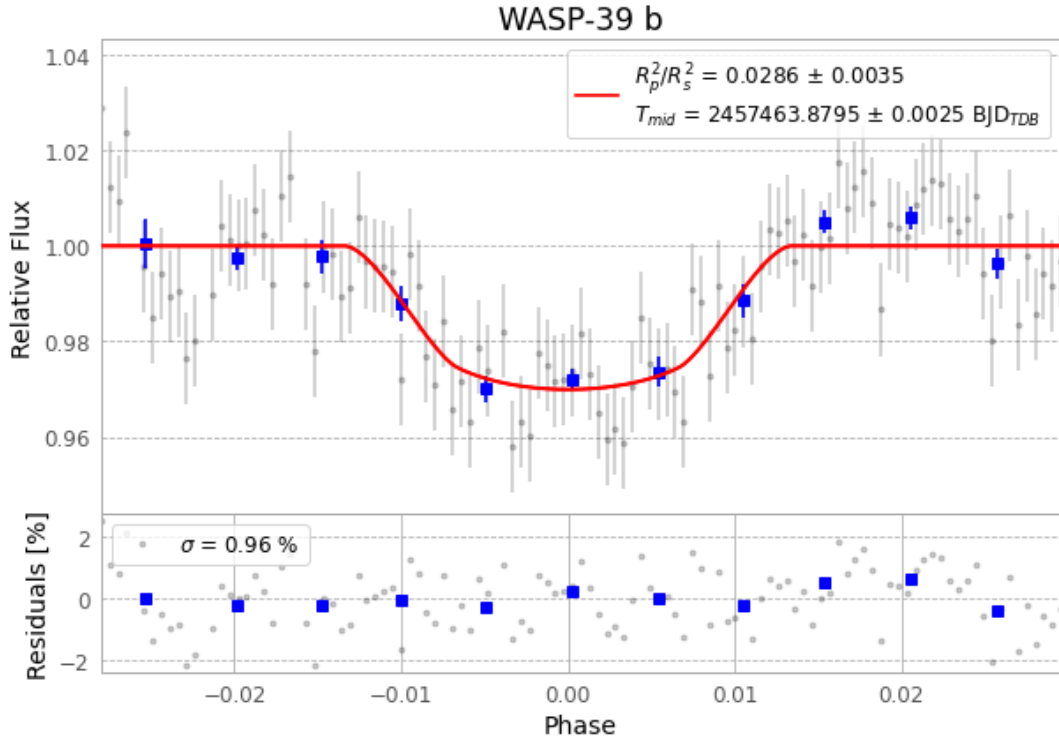


Figure 1 — FinalLightCurve_WASP-39 b_2016-03-15.png (SHA-256 57ad10e577d99a84...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [128, 207], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1de7edea5925d945...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

113 FITS frames (74 MB), WASP-39160316061512.FITS → WASP-39160316115112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-160316.log (c5032c4bbe98...), FinalLightCurve_WASP-39 b_2016-03-15.png (57ad10e577d9...), FinalLightCurve_WASP-39 b_2016-03-15.csv (7d2e506051b9...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 06:47Z.